

Formula sheet: models, parameters, PINN loss

Notation: $x = \log(F/K)$, $\tau = T - t$, $c = C/F$, $w = \sigma_{imp}^2 \tau$, v_s, v_f = slow and fast variance.
Every formula is taken from the code; the file for each is listed in section E.5.

PART A BLACK-SCHOLES

A.1 Equations

$$c(x, \tau, \sigma) = \Phi(d_1) - e^{-x} \Phi(d_2), \quad d_1 = \frac{x}{\sigma\sqrt{\tau}} + \frac{\sigma\sqrt{\tau}}{2}, \quad d_2 = d_1 - \sigma\sqrt{\tau}$$

Currency price $C = Fc$; with $r = q = 0$ (controlled benchmark) $F = S$, so $C = Sc$.

$$\text{vega: } \nu = \frac{\partial c}{\partial \sigma} = \varphi(d_1) \sqrt{\tau}, \quad \varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Implied volatility: invert $C/K = e^x \Phi(d) - \Phi(d - \sqrt{w})$, $d = x/\sqrt{w} + \sqrt{w}/2$, for w by 75 bisection steps on $w \in [10^{-12}, 40]$, then $\sigma_{imp} = \sqrt{w/\tau}$.

Defined only where $\max(e^x - 1, 0) + 10^{-12} < C/K < e^x - 10^{-12}$ - outside it the IV is NaN, which is what produces the gaps in deep out-of-the-money short-dated smiles.

Term-structure variant (BS_TERM / BS_EXPIRY): one σ_j per calibration expiry, total variance interpolated linearly in τ between knots and held flat outside:

$$w(\tau) = \text{interp}(\tau; \{\tau_j, \tau_j \sigma_j^2\}), \quad \sigma(\tau) = \sqrt{w(\tau)/\tau}$$

A.2 Parameters

parameter	what it captures	value or bound used
σ (flat)	one constant volatility for the whole surface	fitted, bounds [0.05, 5]
σ_j (per expiry)	one volatility per calibration expiry	fitted, same bounds; 22-33 knots
$\sigma_{BS, fixed}$	the hard-coded level in fixed-parameter tests	$\sqrt{0.0255} = 15.97\%$

The fixed value is not an independent choice: it is the initial total volatility of the published Double Heston set, so all three models start from the same instantaneous volatility.

PART B SINGLE HESTON

B.1 Equations

$$dS_t = \sqrt{v_t} S_t dW_t^S, \quad dv_t = \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^V, \quad d\langle W^S, W^V \rangle_t = \rho dt$$

Characteristic exponent - the same function used once (Double Heston uses it twice):

$$b = \kappa - \rho \sigma i u, \quad d = \sqrt{b^2 + \sigma^2(u^2 + iu)} \quad \text{with } \Re d \geq 0, \quad g = \frac{b-d}{b+d}$$

$$\psi(u, \tau) = \frac{\kappa \theta}{\sigma^2} \left[(b-d)\tau - 2 \log \frac{1 - g e^{-d\tau}}{1-g} \right] + \frac{b-d}{\sigma^2} \cdot \frac{1 - e^{-d\tau}}{1 - g e^{-d\tau}} v_0$$

$$\varphi(u, \tau) = \exp[\psi(u, \tau)], \quad \text{then the pricing integral of section C.3}$$

B.2 Parameters (5)

symbol	name	what it captures	published	bound
κ	mean reversion	speed volatility returns to normal	8.9814	[0.05, 50]
θ	long-run variance	level volatility drifts back to	0.0409	[0.001, 4]
σ	vol-of-vol	uncertainty of volatility itself	0.2970	[0.01, 10]
ρ	correlation	leverage effect / skew tilt	-0.9621	[-0.99, 0.99]
v_0	initial variance	volatility today	0.0244	[0.001, 4]

Feller on the published set: $2\kappa\theta - \sigma^2 = 0.6465 > 0$ - satisfied.

One κ means ONE timescale: short-dated and long-dated behaviour cannot be matched independently.

PART C DOUBLE HESTON (separable four-shock)

C.1 Dynamics

$$dS_t = \sqrt{v_{s,t} + v_{f,t}} S_t dW_t^S$$

$$dv_{i,t} = \kappa_i(\theta_i - v_{i,t}) dt + \sigma_i \sqrt{v_{i,t}} dW_t^i, \quad d\langle W^S, W^i \rangle_t = \rho_i dt, \quad i \in \{s, f\}$$

Four separate shocks: each variance factor carries its own correlation with the price, and the two variance factors are independent of each other. This is what allows ρ_s and ρ_f to take opposite signs.

C.2 Characteristic function - the two factors multiply

$\psi_i(u, \tau)$ as in B.1, evaluated with $(\kappa_i, \theta_i, \sigma_i, \rho_i, v_{i,0})$

$$\varphi(u, \tau) = \exp[\psi_s(u, \tau) + \psi_f(u, \tau)]$$

The product form is the point of the separable construction: two independent affine factors compose exactly, so a two-factor price costs the same single integral as a one-factor price.

C.3 Pricing integral (forward-normalised, undiscounted; same for Single and Double)

$$c(x, \tau) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{iux} \varphi(u - i, \tau)}{iu} \right] du - e^{-x} \left(\frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{iux} \varphi(u, \tau)}{iu} \right] du \right)$$

Evaluated by Gauss-Laguerre quadrature at 128 nodes and again at 96; the two must agree to 10^{-7} or the point is recomputed by independent adaptive quadrature and the fallback is logged.

C.4 Parameters (10 = 5 per factor)

#	symbol	what it captures	published	bound
1	κ_s	slow mean reversion - half-life 267 days	0.9491	[0.05, 20]
2	θ_s	slow long-run variance (16.0% vol)	0.0257	[0.001, 4]
3	σ_s	slow vol-of-vol	0.0517	[0.01, 10]
4	ρ_s	slow price/variance correlation	+0.7009	[-0.99, 0.99]
5	$v_{s,0}$	slow initial variance (1.7% vol)	0.0003	[0.001, 4]
6	κ_f	fast mean reversion - half-life 24 days	10.7526	gap [0.05, 100]
7	θ_f	fast long-run variance (18.2% vol)	0.0330	[0.001, 4]
8	σ_f	fast vol-of-vol	0.3613	[0.01, 10]
9	ρ_f	fast price/variance correlation	-0.8916	[-0.99, 0.99]
10	$v_{f,0}$	fast initial variance (15.9% vol)	0.0252	[0.001, 4]

Source: Christoffersen, Heston and Jacobs (2009). κ_f is calibrated through the gap $\kappa_f - \kappa_s$, which enforces the slow-first ordering automatically.

Structure: $\kappa_f/\kappa_s = 11.33$ - two genuinely separated timescales; and $\rho_s > 0 > \rho_f$ - opposite-signed correlations, the mechanism behind maturity-dependent skew and the reason the separable four-shock convention is required rather than a restricted single-correlation form.

$$v_{s,0} + v_{f,0} = 0.0255 \Rightarrow 15.97\% \text{ today}, \quad \theta_s + \theta_f = 0.0587 \Rightarrow 24.2\% \text{ long-run}$$

Feller: slow $2\kappa_s\theta_s - \sigma_s^2 = 0.0461 > 0$, fast $0.5791 > 0$ - both satisfied.

Joint disk: $\rho_s^2 + \rho_f^2 = 1.286 > 1$ - the published set fails the stricter repository convention, which is why the market tests use the individual- ρ contract.

C.5 The level scale s - the only number fitted in fixed-parameter experiments

$$\theta_i \mapsto s\theta_i, \quad v_{i,0} \mapsto s v_{i,0}, \quad \sigma_i \mapsto \sqrt{s} \sigma_i, \quad \sigma_{BS} \mapsto \sqrt{s} \sigma_{BS}, \quad s \in [0.7, 3.2]$$

Shifts the volatility LEVEL only; every timescale, correlation and shape is untouched. The range is exactly the range the PINN was trained on.

C.6 Controlled scenario FIXED_TOTAL_TWIST

state	$v_{f,0}$	$v_{s,0}$	total	instantaneous vol	ATM IV at 90 d
FAST-HEAVY	0.035	0.005	0.04	20%	19.95%
SLOW-HEAVY	0.005	0.035	0.04	20%	23.77%

Identical variance today, materially different prices at every maturity, because variance in the fast factor decays 11.3x faster. No single-factor model can produce both from the same starting volatility.

PART D THE PINN (a surrogate for exact Double Heston)

D.1 Output construction - analytic prior plus bounded correction

$$\bar{v}(\tau) = \sum_{i \in \{s, f\}} \left[\theta_i + (v_i - \theta_i) \frac{1 - e^{-\kappa_i \tau}}{\kappa_i \tau} \right] \quad (\text{expected average variance})$$

$$\text{corr} = 1.8 \tanh(\text{head}), \quad \sigma_{\text{imp}} = \sqrt{\bar{v}} e^{\text{corr}}, \quad w = \sigma_{\text{imp}}^2 \tau, \quad C/K = e^x \Phi(d) - \Phi(d - \sqrt{w})$$

Consequences by construction, not by training: the payoff at $\tau = 0$ is exact, and the price cannot leave its no-arbitrage bounds.

D.2 Inputs and features

12 raw inputs: x, v_s, v_f, τ and the 8 structural parameters $\kappa_i, \theta_i, \sigma_i, \rho_i$. Expanded to 24 fixed features (nothing learned), with $z = x/\sqrt{\tau \bar{v} + x^2/64}$ and $\eta_i = \sigma_i/\sqrt{2\kappa_i\theta_i}$:

4 general: $x/0.36, z/8, \tanh(z/1.5), \text{scaled log } \tau$

9 per factor (18): $\log v_i, \log \kappa_i, \log \theta_i, \rho_i, \tanh(\kappa_i \tau), v_i = \tanh(\sigma_i \sqrt{\tau}/\sqrt{\bar{v}}), \rho_i v_i, \tanh(\frac{1}{2} \log(v_i/\theta_i)), 2 \log(1 + \eta_i)/\log 4 - 1$

2 cross-factor: $2v_s/(v_s + v_f) - 1, \tanh(z/1.5) \cdot (\rho_s v_s + \rho_f v_f)$

D.3 The four loss terms

term	definition	what it enforces	points
L_{price}	$\langle (\hat{C}_\theta - C^{\text{teacher}})^2 \rangle$	reproduce the exact Fourier price	100,000 labelled
L_{IV}	$\langle (\hat{\sigma}_\theta - \sigma^{\text{teacher}})^2 \rangle$	accuracy where prices are small	same, valid IV only
L_{PDE}	$\langle \mathcal{R}^2 \rangle$	the two-factor Heston equation	18,000 unlabelled
L_{conv}	$\langle \max(-C, 0)^2 \rangle$	no butterfly arbitrage	same collocation set

$\mathcal{R}$ is the PDE residual in log-total-variance form $\ell = \log w$, normalised by scale = $(v_s + v_f) + \bar{v}$, with $a = \frac{1}{2}x^2 - \frac{1}{8}w^2 - \frac{1}{2}w$; all derivatives by automatic differentiation:

$$\mathcal{R} = \frac{1}{\text{scale}} \left[w \ell_\tau - \frac{v_s + v_f}{2} D_x - \sum_i \rho_i \sigma_i v_i D_{xv_i} - \frac{1}{2} \sum_i \sigma_i^2 v_i D_{v_i v_i} - \sum_i \kappa_i (\theta_i - v_i) w \ell_{v_i} \right]$$

$$D_x = 2 + 2(\frac{w}{2} - x) \ell_x + a \ell_x^2 + w(\ell_{xx} + \ell_x^2) - w \ell_x$$

$$D_{xv_i} = (\frac{w}{2} - x) \ell_{v_i} + a \ell_x \ell_{v_i} + w(\ell_{xv_i} + \ell_x \ell_{v_i}), \quad D_{v_i v_i} = a \ell_{v_i}^2 + w(\ell_{v_i v_i} + \ell_{v_i}^2)$$

$$C = (1 - \frac{1}{2}x \ell_x)^2 - \frac{1}{4}w \ell_x^2 - \frac{w^2 \ell_x^2}{16} + \frac{1}{2}w(\ell_{xx} + \ell_x^2) \quad (\text{convexity diagnostic})$$

D.4 The total loss function

Each supervised term is divided by the square of its own acceptance tolerance, so no term dominates merely because of its units:

$$L = \frac{L_{price}}{(2 \times 10^{-5})^2} + w_{IV} \frac{L_{IV}}{(0.002)^2} + w_{PDE} \cdot 0.1 \cdot \frac{L_{PDE}}{(0.01)^2} + w_{conv} \cdot 0.1 \cdot L_{conv}$$

weights	locked v4	improved v5
$w_{IV}, w_{PDE}, w_{conv}$	fixed at 1	updated every 100 steps by gradient balancing
$w_i \leftarrow 0.9 w_i + 0.1 \cdot \min \left(\frac{\ \nabla_{\theta} L_{price}\ }{\ \nabla_{\theta} L_i\ }, 10^3 \right)$		

Learning-rate annealing of Wang, Teng and Perdikaris: if one term produces gradients orders of magnitude larger than the price term it silently dominates; this rebalances toward the ratio of gradient norms, smoothed 90/10.

D.5 How the loss is minimised

stage	setting
Adam	40,000 steps; lr $10^{-3} \rightarrow 10^{-5}$ cosine; batch 512 labelled + 128 collocation
L-BFGS	300 steps, strong-Wolfe line search, history 30, on 4,096 labels + 512 collocation
seeds	17 and 43 trained independently; the two predicted prices are averaged
collocation (v5)	half of the 18,000 points redrawn every 2,000 steps, $p_j \propto \mathcal{R}_j / \overline{\mathcal{R}} + 1$
checkpointing	every 2,000 steps, so an interrupted run costs minutes not hours

Acceptance gates, frozen before training - all four must pass on the development split:

$$\text{price RMSE} \leq 2 \times 10^{-5}, \quad P_{95} \leq 5 \times 10^{-5}, \quad \max \leq 2 \times 10^{-4}, \quad \text{IV RMSE} \leq 0.002 \quad (0.2 \text{ vol pts})$$

D.6 Architecture and measured effect

	locked v4	improved v5
body	plain stack: 5 tanh layers, width 256	5 gated blocks, width 256
gating	none	$h \leftarrow (1 - Z_t) \odot U + Z_t \odot V$
activation	fixed tanh	$\tanh(a_t z)$, a_t learned per block
parameters	269,825	282,632 (+4.7%)
price RMSE vs exact DH	1.062e-5	6.610e-6 (1.61x better)
IV RMSE (vol points)	0.1175	0.0738 (1.59x better)
worst-case price error	1.523e-4	5.938e-5 (2.56x better)
PDE residual	0.0309	0.0558 (1.8x worse)
inference latency	14.2 $\mu\text{s}/\text{quote}$	24.8 $\mu\text{s}/\text{quote}$ (1.7x slower)

Measured once on an untouched 16,384-point test set. The improvement is to the SOLVER: it changed real SPX market error by 0.003 vol points out of 5.8, because network error is about 1% of model error.

PART E CONDITIONS, DATA RANGES, CODE MAP

E.1 Parameter admissibility

condition	formula	where enforced
positivity	$\kappa_i, \theta_i, \sigma_i, v_{0,i} > 0$	always
correlation bounds	$-1 < \rho_i < 1$	always
slow-first ordering	$\kappa_s < \kappa_f$	always (removes the label swap)
Feller (strict variant)	$2\kappa_i\theta_i - \sigma_i^2 > 0$ via $\sigma_i = \eta_i\sqrt{2\kappa_i\theta_i}$	feller_strict only
joint correlation disk	$\rho_s^2 + \rho_f^2 < 1$	reference contract, NOT market tests

E.2 Numerical and no-arbitrage conditions

$$|c_{128} - c_{96}| \leq 10^{-7}, \quad c \geq \max(1 - e^{-x}, 0) - 10^{-9}, \quad c \leq 1 + 10^{-9}$$

$$\partial C / \partial S \in [0, 1], \quad \partial^2 C / \partial S^2 \geq 0, \quad \partial^2 C / \partial K^2 \geq 0, \quad \partial w / \partial \tau \geq 0, \quad \max(S - K, 0) \leq C \leq S$$

Checked on all 30 controlled curves: zero violations.

E.3 Calibration objective and PINN training domain

$$\min \left\langle \left(\frac{C_{\text{model}} - C_{\text{market}}}{\max(v, 10^{-4})} \right)^2 \right\rangle \quad \text{differential evolution (popsize x6, 25 iters), then 12 local starts}$$

variable	PINN training range	sampling
$x = \log(F/K)$	$[-0.36, +0.36]$ i.e. $S/K \in [0.698, 1.433]$	uniform
τ	7 days to 2 years	uniform in $\log \tau$
v_s	$[1.5 \times 10^{-4}, 0.18]$ i.e. 1.2% - 42% vol	uniform in $\log v_s$
v_f	$[1.0 \times 10^{-3}, 0.22]$ i.e. 3.2% - 47% vol	uniform in $\log v_f$
level scale s	$[0.7, 3.2]$	uniform
datasets	train 100k, collocation 18k, dev 8,192, untouched test 16,384 5-D Latin hypercube	

E.4 Market data filters

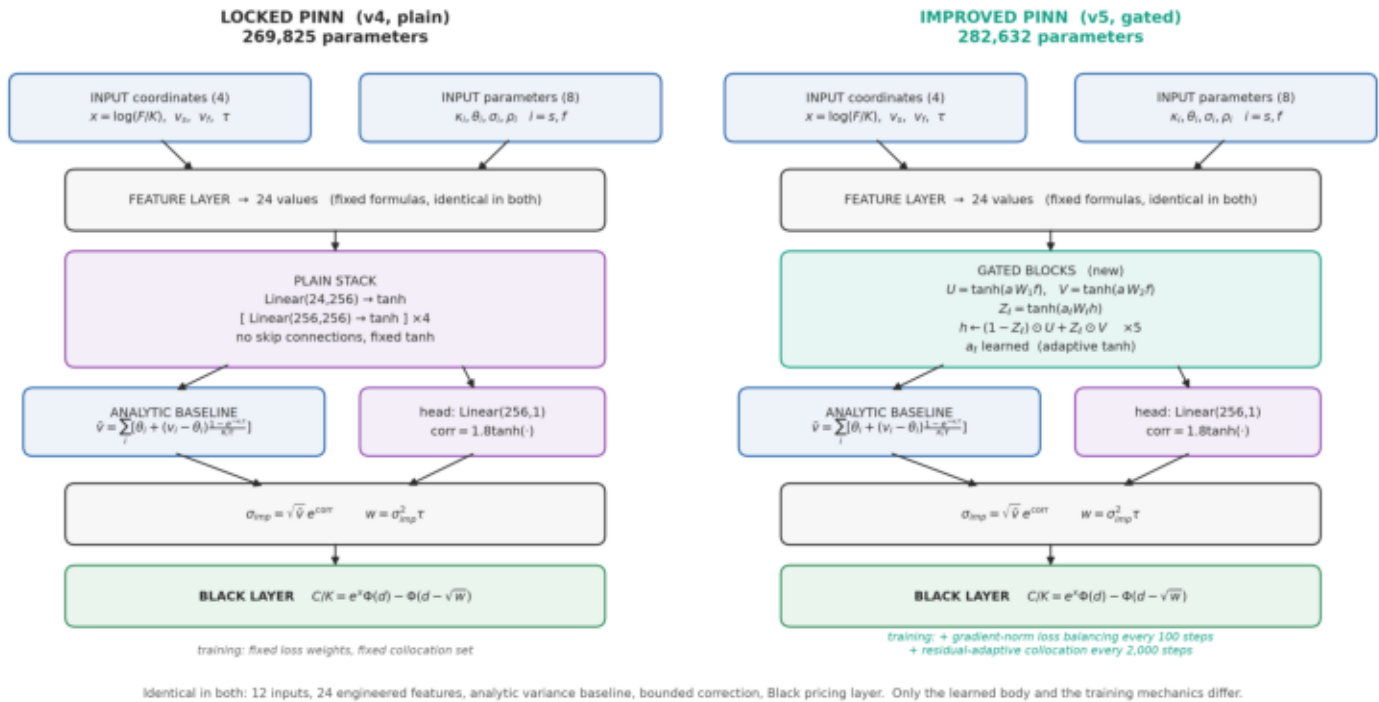
	crypto (Deribit)	equity / index (CBOE)
maturity	3 - 400 days	7 - 730 days
moneyness	$ x \leq 1.0$	$ x \leq 0.36$ (the PINN domain)
implied vol	0.10 - 3.00	0.02 - 3.00
quotes / expiries	≥ 3 per expiry, ≥ 5 expiries (BTC, ETH)	≥ 6 per expiry, ≥ 6 expiries
forward	solved per trade from (price, IV)	parity regression, $R^2 \geq 0.999$

E.5 Where each equation lives

equation	file
Black-76 price, vega, implied vol	experiments/btc_multifactor_v1/engine.py
IV inversion and validity window	src/mentor_dh_pinn/regular_pinn_data.py
characteristic exponent ψ	src/double_heston_reference.py::factor_exponent
pricing integral	experiments/btc_multifactor_v1/engine.py::Grid
admissibility, Feller, disk	engine.py::admissible, src/constraints.py
baseline, features, correction, PDE residual	src/mentor_dh_pinn/regular_pinn_torch.py
loss, gradient balancing, RAD sampling	experiments/dh_pinn_v5/train.py
published parameters, domains	experiments/nifty_multifactor_v4/config.json
calibration bounds and objective	experiments/btc_multifactor_v1/config.json

D.7 Block architecture, locked v4 and improved v5

Double Heston PINN: prior-plus-correction architecture, before and after



Identical in both: 12 inputs, 24 engineered features, analytic variance baseline, bounded correction and the Black pricing layer. Only the learned body and the training mechanics differ.

Summary of the three model sections

	free parameters	smile?	two horizons?	held-out IWM error
Black-Scholes (per expiry)	22	no	partly, by expiry	5.39 vol pts
Single Heston	5	yes	no - one timescale	1.89 vol pts
Double Heston	10	yes	yes - fast and slow	0.58 vol pts
DH-PINN (surrogate)	reproduces DH	inherits DH	inherits DH	7.5e-4 vs exact DH

Companion document: HOW_IT_WORKS.pdf - the same material explained in plain language.